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**42136 - Large Scale Optimization using Decomposition
Assignment Part II**

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1 Exercise 1

1.1 Visualization

In this task, we analyze the Integer Program (IP) with objective function

$$\max x + 2y$$

subject to a set of constraints. The feasible region of the IP is given by the intersection of all the constraints, restricted to integer-valued points. The LP relaxation is obtained by replacing the integrality constraint $(x, y) \in \mathbb{Z}^2$ with $(x, y) \in \mathbb{R}^2$.

Figure 1 illustrates the feasible regions defined by the constraint sets. The blue region corresponds to the feasible set induced by the A1 constraints, while the red region represents the feasible set defined by the general (A0) constraints.

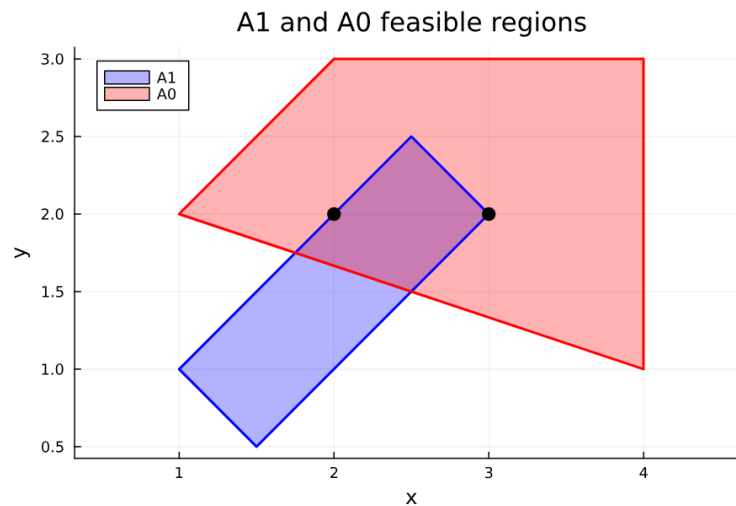


Figure 1: Feasible regions

The feasible region of the full problem is given by the intersection of these two regions. The LP relaxation corresponds to this intersection as a convex polyhedron, whereas the IP feasible solutions are the integer points contained within this region. Selected integer feasible points are highlighted in the figure.

1.2 Dantzig–Wolfe Relaxation

We consider the set X_1 , defined as the convex hull of the integer feasible solutions satisfying the A1 constraints.

Extreme Points of X_1 : From the feasible region defined by the A1 constraints and integrality, the extreme points are:

$$(1, 1), \quad (2, 1), \quad (2, 2), \quad (3, 2).$$

Any point $(x, y) \in X_1$ can therefore be written as a convex combination of these points:

$$(x, y) = \lambda_1(1, 1) + \lambda_2(2, 1) + \lambda_3(2, 2) + \lambda_4(3, 2), \text{ with}$$

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1, \quad \lambda_i \geq 0.$$

Dantzig–Wolfe Reformulation Substituting the convex combination into the objective function:

$$x + 2y = 3\lambda_1 + 4\lambda_2 + 6\lambda_3 + 7\lambda_4,$$

we obtain the following linear program:

$$\begin{aligned} \max \quad & 3\lambda_1 + 4\lambda_2 + 6\lambda_3 + 7\lambda_4 \\ \text{s.t.} \quad & \lambda_1 + \lambda_2 + 2\lambda_3 + 2\lambda_4 \leq 3 \\ & \lambda_1 + 2\lambda_2 + 2\lambda_3 + 3\lambda_4 \leq 4 \\ & 4\lambda_1 + 5\lambda_2 + 8\lambda_3 + 9\lambda_4 \geq 7 \\ & \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1 \\ & \lambda_i \geq 0 \quad (i = 1, \dots, 4). \end{aligned}$$

Solution of the Relaxation: Solving the Dantzig–Wolfe relaxation yields $\lambda = (0, 0, 0, 1)$, which corresponds to the point $(x, y) = (3, 2)$ with objective value $z = 7$.

Comparison with IP and LP Solutions From Figure 2, we observe that:

- The optimal IP and LP solution is $(2, 3)$ with objective value 8.
- The Dantzig–Wolfe relaxation yields the solution $(3, 2)$ with objective value 7.

Thus, the Dantzig–Wolfe relaxation provides a lower bound on the optimal value of the original problem.

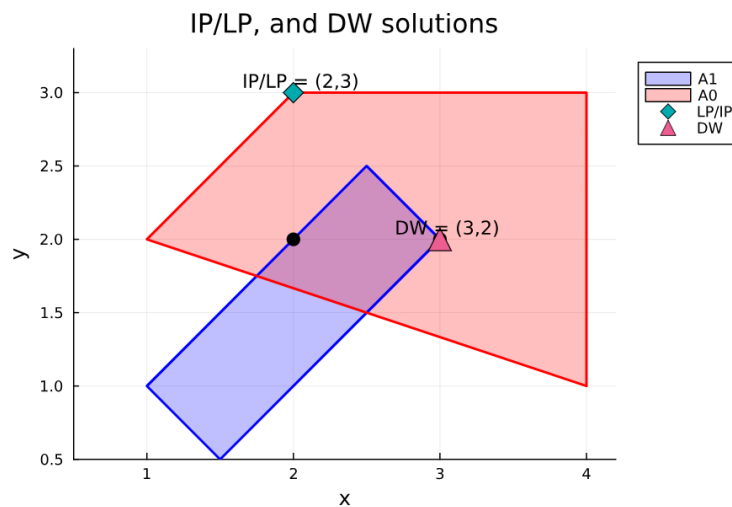


Figure 2: Comparison of IP/LP solution and Dantzig–Wolfe relaxation

2 Exercise 2

2.1 MIP Model and Solution

The model above was implemented directly in Julia using JuMP and GLPK, with all four constraint types included. The MIP solver found the following optimal integer solution:

Table 1: Optimal MIP solution

Machine	Jobs Assigned	Profit
Machine 1	{2, 3}	$7 + 3 = 10$
Machine 2	{4, 5}	$7 + 9 = 16$
Machine 3	{1}	7

The optimal objective value is:

$$Z_{\text{MIP}} = p_{1,2} + p_{1,3} + p_{2,4} + p_{2,5} + p_{3,1} = 7 + 3 + 7 + 9 + 7 = \mathbf{33}$$

2.2 Dantzig-Wolfe Decomposition via Column Generation

Constraint (1) is kept in the **master problem**. Constraints (2), (3) and (4) are **convexified** into one subproblem per machine i . The feasible integer solutions of subproblem i define a set of extreme points $\{x_i^k\}_{k \in \mathcal{K}_i}$.

Decomposition Structure

Introducing variables $\lambda_i^k \geq 0$ (the weight of extreme point k for machine i), the DW master LP reads:

$$\max \sum_{i \in I} \sum_{k \in \mathcal{K}_i} \left(\sum_{j \in J} p_{ij} x_{ij}^k \right) \lambda_i^k \quad (1)$$

$$\text{s.t.} \quad \sum_{i \in I} \sum_{k \in \mathcal{K}_i} x_{ij}^k \lambda_i^k = 1 \quad \forall j \in J \quad [\mu_j] \quad (2)$$

$$\sum_{k \in \mathcal{K}_i} \lambda_i^k = 1 \quad \forall i \in I \quad [\sigma_i] \quad (3)$$

$$\lambda_i^k \geq 0 \quad \forall i \in I, k \in \mathcal{K}_i \quad (4)$$

where μ_j and σ_i are the dual variables associated with the assignment and convexity constraints, respectively.

Pricing Subproblem

For each machine i , the pricing subproblem identifies the most improving column (extreme point) to add to the master:

$$\max \sum_{j \in J} (p_{ij} - \mu_j) x_j - \sigma_i \quad (5)$$

$$\text{s.t.} \quad \sum_{j \in J} w_{ij} x_j \leq c_i \quad (6)$$

$$x_a + x_b \leq 1 \quad \forall (a, b) \in K \quad (7)$$

$$x_j \in \{0, 1\} \quad \forall j \in J \quad (8)$$

A column is added to the master if the reduced cost is strictly positive, i.e., the subproblem objective value $> \varepsilon$ (here $\varepsilon = 10^{-5}$).

Results

The column generation converged after **8 iterations**, $Z_{\text{DW}} = \mathbf{33.0}$:

Table 2: Active columns in the DW solution

Column	Machine	Jobs	λ
λ_2^6	2	{4, 5}	1.0
λ_1^{11}	1	{2, 3}	1.0
λ_3^{21}	3	{1}	1.0

2.3 Explicit LP Relaxation of the DW Reformulation

Instead of generating columns on the fly, all extreme points of each subproblem are **explicitly enumerated** by iterating over all $2^{|J|} = 32$ binary combinations per machine and retaining only those satisfying Constraint (2) and (3).

The resulting extreme points populate the full DW master LP from the outset, which is then solved directly as a continuous LP ($\lambda_i^k \geq 0$).

Results

The explicit DW LP yields:

$$Z_{\text{LP}} = \mathbf{33.0}$$

This matches the result of Task 2.2 exactly, confirming the theoretical equivalence: both approaches solve the **same LP**. Column generation (Task 2.2) finds the optimum implicitly by generating only the columns needed to prove optimality, while Task 2.3 enumerates all columns upfront and solves the full LP directly.